Solutions of the Z3_RR NCq3 chain

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A következőképp kell olvasni az eredményeket:

1. A Z3_RR elnevezés a ciklikus bázistranszformációra (Z3) és tükrözésre (R) véve szimmetrikus, valós (R) operátorokat jelenti.

2. h_tilde: Az első egy olyan klasszifikáció, ahol nem volt az extra $\tilde{h}$ tag a $\mathcal{Q}_5$ töltésben, míg a diag-gauge klasszifikációban egy
$$\tilde{h}_j = \sum_{a,c=0}^{2} \gamma_{ac} P_j^a P_{j+1}^c \tag{1}$$
diagonális tag adódik ehhez.

3. Megoldások felcímkézése: szerepel a megoldás száma (no. n), a család neve (ez esetben Z3_RR, valamint a $\tilde{h}$ fajtája), a minta, vagyis a nem diagonális h^{ab} blokkok labeljei, a chart száma az adott mintában (ez a konstans szorzó, amit említettem múltkor), valamint az adott mintán belül a megtalált megoldás száma.

Mind a 9 blokk listázva, címkézve van, megfelelnek az ansatz formájának. A diagonális gauge esetén a 9 γ_{ac} paraméter van listázva, a hozzájuk tartozó operátorok csak projektorok, ugyebár.

1 Z3_RR

Genuine Solution no. 0 for family Z3_RR_None with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 0:

$$h_0^{00} = \begin{pmatrix} u_{22}^{22} & \left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}+1\right)\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & -\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} \\ \left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}+1\right)\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & u_{22}^{11} & 1 \\ -\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & 1 & -\frac{2u_{22}^{11}u_{22}^{12}-3u_{22}^{11}u_{22}^{22}+3u_{22}^{11}-4\left(u_{22}^{12}\right)^2+8u_{22}^{12}u_{22}^{22}-8u_{22}^{12}-3\left(u_{22}^{22}\right)^2+6u_{22}^{22}-4}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1} \end{pmatrix}$$

$$h_0^{01} = \begin{pmatrix} -u_{22}^{12}+2u_{22}^{22}-2 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22}-1 \end{pmatrix}$$

$$h_0^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22}-1 & 0 \\ 0 & 0 & -u_{22}^{12}+2u_{22}^{22}-2 \end{pmatrix}$$

$$h_0^{10} = \begin{pmatrix} -u_{22}^{12}+2u_{22}^{22}-2 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22}-1 \end{pmatrix}$$

$$h_0^{11} = \begin{pmatrix} -\frac{2u_{22}^{11}u_{22}^{12}-3u_{22}^{11}u_{22}^{22}+3u_{22}^{11}-4\left(u_{22}^{12}\right)^2+8u_{22}^{12}u_{22}^{22}-8u_{22}^{12}-3\left(u_{22}^{22}\right)^2+6u_{22}^{22}-4}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1} & -\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & 1 \\ -\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & u_{22}^{22} & \left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}+1\right)\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} \\ 1 & \left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}+1\right)\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & u_{22}^{11} \end{pmatrix}$$

$$h_0^{12} = \begin{pmatrix} u_{22}^{22}-1 & 0 & 0 \\ 0 & -u_{22}^{12}+2u_{22}^{22}-2 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_0^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22}-1 & 0 \\ 0 & 0 & -u_{22}^{12}+2u_{22}^{22}-2 \end{pmatrix}$$

$$h_0^{21} = \begin{pmatrix} u_{22}^{22}-1 & 0 & 0 \\ 0 & -u_{22}^{12}+2u_{22}^{22}-2 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_0^{22} = \begin{pmatrix} u_{22}^{11} & 1 & \left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}+1\right)\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} \\ 1 & -\frac{2u_{22}^{11}u_{22}^{12}-3u_{22}^{11}u_{22}^{22}+3u_{22}^{11}-4\left(u_{22}^{12}\right)^2+8u_{22}^{12}u_{22}^{22}-8u_{22}^{12}-3\left(u_{22}^{22}\right)^2+6u_{22}^{22}-4}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1} & -\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} \\ \left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}+1\right)\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & -\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & u_{22}^{22} \end{pmatrix}$$

Genuine Solution no. 1 for family Z3_RR_None with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 1:

$$h_1^{00} = \begin{pmatrix} u_{22}^{22} & \sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}}\left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}-1\right) & -\sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}} \\ \sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}}\left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}-1\right) & u_{22}^{11} & 1 \\ -\sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}} & 1 & -\frac{2u_{22}^{11}u_{22}^{12}-3u_{22}^{11}u_{22}^{22}-3u_{22}^{11}-4\left(u_{22}^{12}\right)^2+8u_{22}^{12}u_{22}^{22}+8u_{22}^{12}-3\left(u_{22}^{22}\right)^2-6u_{22}^{22}-4}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1} \end{pmatrix}$$

$$h_1^{01} = \begin{pmatrix} -u_{22}^{12}+2u_{22}^{22}+2 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22}+1 \end{pmatrix}$$

$$h_1^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22}+1 & 0 \\ 0 & 0 & -u_{22}^{12}+2u_{22}^{22}+2 \end{pmatrix}$$

$$h_1^{10} = \begin{pmatrix} -u_{22}^{12}+2u_{22}^{22}+2 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22}+1 \end{pmatrix}$$

$$h_1^{11} = \begin{pmatrix} -\frac{2u_{22}^{11}u_{22}^{12}-3u_{22}^{11}u_{22}^{22}-3u_{22}^{11}-4\left(u_{22}^{12}\right)^2+8u_{22}^{12}u_{22}^{22}+8u_{22}^{12}-3\left(u_{22}^{22}\right)^2-6u_{22}^{22}-4}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1} & -\sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}} & 1 \\ -\sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}} & u_{22}^{22} & \sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}}\left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}-1\right) \\ 1 & \sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}}\left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}-1\right) & u_{22}^{11} \end{pmatrix}$$

$$h_1^{12} = \begin{pmatrix} u_{22}^{22}+1 & 0 & 0 \\ 0 & -u_{22}^{12}+2u_{22}^{22}+2 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22}+1 & 0 \\ 0 & 0 & -u_{22}^{12}+2u_{22}^{22}+2 \end{pmatrix}$$

$$h_1^{21} = \begin{pmatrix} u_{22}^{22}+1 & 0 & 0 \\ 0 & -u_{22}^{12}+2u_{22}^{22}+2 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1^{22} = \begin{pmatrix} u_{22}^{11} & 1 & \sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}}\left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}-1\right) \\ 1 & -\frac{2u_{22}^{11}u_{22}^{12}-3u_{22}^{11}u_{22}^{22}-3u_{22}^{11}-4\left(u_{22}^{12}\right)^2+8u_{22}^{12}u_{22}^{22}+8u_{22}^{12}-3\left(u_{22}^{22}\right)^2-6u_{22}^{22}-4}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1} & -\sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}} \\ \sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}}\left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}-1\right) & -\sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}} & u_{22}^{22} \end{pmatrix}$$

Genuine Solution no. 3 for family Z3_RR_None with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 3:

$$h_3^{00} = \begin{pmatrix} u_{22}^{22} & 1 & 1 \\ 1 & 2u_{22}^{12} - u_{22}^{22} + 4 & 1 \\ 1 & 1 & -2u_{22}^{12} + 3u_{22}^{22} - 4 \end{pmatrix}$$

$$h_3^{01} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} - 2 \end{pmatrix}$$

$$h_3^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} - 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 \end{pmatrix}$$

$$h_3^{10} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} - 2 \end{pmatrix}$$

$$h_3^{11} = \begin{pmatrix} -2u_{22}^{12} + 3u_{22}^{22} - 4 & 1 & 1 \\ 1 & u_{22}^{22} & 1 \\ 1 & 1 & 2u_{22}^{12} - u_{22}^{22} + 4 \end{pmatrix}$$

$$h_3^{12} = \begin{pmatrix} u_{22}^{22} - 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_3^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} - 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 \end{pmatrix}$$

$$h_3^{21} = \begin{pmatrix} u_{22}^{22} - 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_3^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} + 4 & 1 & 1 \\ 1 & -2u_{22}^{12} + 3u_{22}^{22} - 4 & 1 \\ 1 & 1 & u_{22}^{22} \end{pmatrix}$$

Genuine Solution no. 4 for family Z3_RR_None with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 4:

$$h_4^{00} = \begin{pmatrix} u_{22}^{22} & -1 & 1 \\ -1 & 2u_{22}^{12} - u_{22}^{22} - 4 & 1 \\ 1 & 1 & -2u_{22}^{12} + 3u_{22}^{22} + 4 \end{pmatrix}$$

$$h_4^{01} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} + 2 \end{pmatrix}$$

$$h_4^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} + 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 \end{pmatrix}$$

$$h_4^{10} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} + 2 \end{pmatrix}$$

$$h_4^{11} = \begin{pmatrix} -2u_{22}^{12} + 3u_{22}^{22} + 4 & 1 & 1 \\ 1 & u_{22}^{22} & -1 \\ 1 & -1 & 2u_{22}^{12} - u_{22}^{22} - 4 \end{pmatrix}$$

$$h_4^{12} = \begin{pmatrix} u_{22}^{22} + 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_4^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} + 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 \end{pmatrix}$$

$$h_4^{21} = \begin{pmatrix} u_{22}^{22} + 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_4^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} - 4 & 1 & -1 \\ 1 & -2u_{22}^{12} + 3u_{22}^{22} + 4 & 1 \\ -1 & 1 & u_{22}^{22} \end{pmatrix}$$

Genuine Solution no. 5 for family Z3_RR_None with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 5:

$$h_5^{00} = \begin{pmatrix} u_{22}^{22} & -\frac{\sqrt{2+2\sqrt{5}}}{2} & -\sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} \\ -\frac{\sqrt{2+2\sqrt{5}}}{2} & 2u_{22}^{12} - u_{22}^{22} + 2 + \sqrt{5} & 1 \\ -\sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & 1 & -2u_{22}^{12} + 3u_{22}^{22} - \frac{3\sqrt{5}}{2} - \frac{1}{2} \end{pmatrix}$$

$$h_5^{01} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} - \frac{5\sqrt{5}}{4} - \frac{5}{4} & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} - \frac{3\sqrt{5}}{4} - \frac{3}{4} \end{pmatrix}$$

$$h_5^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} - \frac{3\sqrt{5}}{4} - \frac{3}{4} & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} - \frac{5\sqrt{5}}{4} - \frac{5}{4} \end{pmatrix}$$

$$h_5^{10} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} - \frac{5\sqrt{5}}{4} - \frac{5}{4} & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} - \frac{3\sqrt{5}}{4} - \frac{3}{4} \end{pmatrix}$$

$$h_5^{11} = \begin{pmatrix} -2u_{22}^{12} + 3u_{22}^{22} - \frac{3\sqrt{5}}{2} - \frac{1}{2} & -\sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & 1 \\ -\sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & u_{22}^{22} & -\frac{\sqrt{2+2\sqrt{5}}}{2} \\ 1 & -\frac{\sqrt{2+2\sqrt{5}}}{2} & 2u_{22}^{12} - u_{22}^{22} + 2 + \sqrt{5} \end{pmatrix}$$

$$h_5^{12} = \begin{pmatrix} u_{22}^{22} - \frac{3\sqrt{5}}{4} - \frac{3}{4} & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} - \frac{5\sqrt{5}}{4} - \frac{5}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_5^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} - \frac{3\sqrt{5}}{4} - \frac{3}{4} & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} - \frac{5\sqrt{5}}{4} - \frac{5}{4} \end{pmatrix}$$

$$h_5^{21} = \begin{pmatrix} u_{22}^{22} - \frac{3\sqrt{5}}{4} - \frac{3}{4} & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} - \frac{5\sqrt{5}}{4} - \frac{5}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_5^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} + 2 + \sqrt{5} & 1 & -\frac{\sqrt{2+2\sqrt{5}}}{2} \\ 1 & -2u_{22}^{12} + 3u_{22}^{22} - \frac{3\sqrt{5}}{2} - \frac{1}{2} & -\sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} \\ -\frac{\sqrt{2+2\sqrt{5}}}{2} & -\sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & u_{22}^{22} \end{pmatrix}$$

Genuine Solution no. 6 for family Z3_RR_None with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 6:

$$h_6^{00} = \begin{pmatrix} u_{22}^{22} & -\frac{\sqrt{2+2\sqrt{5}}}{2} & \sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} \\ -\frac{\sqrt{2+2\sqrt{5}}}{2} & 2u_{22}^{12} - u_{22}^{22} - \sqrt{5} - 2 & 1 \\ \sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & 1 & -2u_{22}^{12} + 3u_{22}^{22} + \frac{1}{2} + \frac{3\sqrt{5}}{2} \end{pmatrix}$$

$$h_6^{01} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} + \frac{5}{4} + \frac{5\sqrt{5}}{4} & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} + \frac{3}{4} + \frac{3\sqrt{5}}{4} \end{pmatrix}$$

$$h_6^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} + \frac{3}{4} + \frac{3\sqrt{5}}{4} & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} + \frac{5}{4} + \frac{5\sqrt{5}}{4} \end{pmatrix}$$

$$h_6^{10} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} + \frac{5}{4} + \frac{5\sqrt{5}}{4} & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} + \frac{3}{4} + \frac{3\sqrt{5}}{4} \end{pmatrix}$$

$$h_6^{11} = \begin{pmatrix} -2u_{22}^{12} + 3u_{22}^{22} + \frac{1}{2} + \frac{3\sqrt{5}}{2} & \sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & 1 \\ \sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & u_{22}^{22} & -\frac{\sqrt{2+2\sqrt{5}}}{2} \\ 1 & -\frac{\sqrt{2+2\sqrt{5}}}{2} & 2u_{22}^{12} - u_{22}^{22} - \sqrt{5} - 2 \end{pmatrix}$$

$$h_6^{12} = \begin{pmatrix} u_{22}^{22} + \frac{3}{4} + \frac{3\sqrt{5}}{4} & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} + \frac{5}{4} + \frac{5\sqrt{5}}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_6^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} + \frac{3}{4} + \frac{3\sqrt{5}}{4} & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} + \frac{5}{4} + \frac{5\sqrt{5}}{4} \end{pmatrix}$$

$$h_6^{21} = \begin{pmatrix} u_{22}^{22} + \frac{3}{4} + \frac{3\sqrt{5}}{4} & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} + \frac{5}{4} + \frac{5\sqrt{5}}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_6^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} - \sqrt{5} - 2 & 1 & -\frac{\sqrt{2+2\sqrt{5}}}{2} \\ 1 & -2u_{22}^{12} + 3u_{22}^{22} + \frac{1}{2} + \frac{3\sqrt{5}}{2} & \sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} \\ -\frac{\sqrt{2+2\sqrt{5}}}{2} & \sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & u_{22}^{22} \end{pmatrix}$$

Genuine Solution no. 7 for family Z3_RR_None with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 7:

$$h_7^{00} = \begin{pmatrix} u_{22}^{22} & -1 & -1 \\ -1 & 2u_{22}^{12} - u_{22}^{22} + 4 & 1 \\ -1 & 1 & -2u_{22}^{12} + 3u_{22}^{22} - 4 \end{pmatrix}$$

$$h_7^{01} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} - 2 \end{pmatrix}$$

$$h_7^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} - 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 \end{pmatrix}$$

$$h_7^{10} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} - 2 \end{pmatrix}$$

$$h_7^{11} = \begin{pmatrix} -2u_{22}^{12} + 3u_{22}^{22} - 4 & -1 & 1 \\ -1 & u_{22}^{22} & -1 \\ 1 & -1 & 2u_{22}^{12} - u_{22}^{22} + 4 \end{pmatrix}$$

$$h_7^{12} = \begin{pmatrix} u_{22}^{22} - 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_7^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} - 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 \end{pmatrix}$$

$$h_7^{21} = \begin{pmatrix} u_{22}^{22} - 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_7^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} + 4 & 1 & -1 \\ 1 & -2u_{22}^{12} + 3u_{22}^{22} - 4 & -1 \\ -1 & -1 & u_{22}^{22} \end{pmatrix}$$

Genuine Solution no. 8 for family Z3_RR_None with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 8:

$$h_8^{00} = \begin{pmatrix} u_{22}^{22} & 1 & -1 \\ 1 & 2u_{22}^{12} - u_{22}^{22} - 4 & 1 \\ -1 & 1 & -2u_{22}^{12} + 3u_{22}^{22} + 4 \end{pmatrix}$$

$$h_8^{01} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} + 2 \end{pmatrix}$$

$$h_8^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} + 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 \end{pmatrix}$$

$$h_8^{10} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} + 2 \end{pmatrix}$$

$$h_8^{11} = \begin{pmatrix} -2u_{22}^{12} + 3u_{22}^{22} + 4 & -1 & 1 \\ -1 & u_{22}^{22} & 1 \\ 1 & 1 & 2u_{22}^{12} - u_{22}^{22} - 4 \end{pmatrix}$$

$$h_8^{12} = \begin{pmatrix} u_{22}^{22} + 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_8^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} + 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 \end{pmatrix}$$

$$h_8^{21} = \begin{pmatrix} u_{22}^{22} + 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_8^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} - 4 & 1 & 1 \\ 1 & -2u_{22}^{12} + 3u_{22}^{22} + 4 & -1 \\ 1 & -1 & u_{22}^{22} \end{pmatrix}$$

Genuine Solution no. 9 for family Z3_RR_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=2, comp 0:

$$h_9^{00} = \begin{pmatrix} u_{22}^{22} & 0 & 0 \\ 0 & u_{22}^{11} & 0 \\ 0 & 0 & -u_{22}^{11} + 2u_{22}^{22} \end{pmatrix}$$

$$h_9^{01} = \begin{pmatrix} -u_{22}^{11} + u_{22}^{12} + u_{22}^{22} & 1 & 0 \\ 1 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} \end{pmatrix}$$

$$h_9^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 1 \\ 0 & u_{22}^{22} & 0 \\ 1 & 0 & -u_{22}^{11} + u_{22}^{12} + u_{22}^{22} \end{pmatrix}$$

$$h_9^{10} = \begin{pmatrix} -u_{22}^{11} + u_{22}^{12} + u_{22}^{22} & 1 & 0 \\ 1 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} \end{pmatrix}$$

$$h_9^{11} = \begin{pmatrix} -u_{22}^{11} + 2u_{22}^{22} & 0 & 0 \\ 0 & u_{22}^{22} & 0 \\ 0 & 0 & u_{22}^{11} \end{pmatrix}$$

$$h_9^{12} = \begin{pmatrix} u_{22}^{22} & 0 & 0 \\ 0 & -u_{22}^{11} + u_{22}^{12} + u_{22}^{22} & 1 \\ 0 & 1 & u_{22}^{12} \end{pmatrix}$$

$$h_9^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 1 \\ 0 & u_{22}^{22} & 0 \\ 1 & 0 & -u_{22}^{11} + u_{22}^{12} + u_{22}^{22} \end{pmatrix}$$

$$h_9^{21} = \begin{pmatrix} u_{22}^{22} & 0 & 0 \\ 0 & -u_{22}^{11} + u_{22}^{12} + u_{22}^{22} & 1 \\ 0 & 1 & u_{22}^{12} \end{pmatrix}$$

$$h_9^{22} = \begin{pmatrix} u_{22}^{11} & 0 & 0 \\ 0 & -u_{22}^{11} + 2u_{22}^{22} & 0 \\ 0 & 0 & u_{22}^{22} \end{pmatrix}$$

Genuine Solution no. 10 for family Z3_RR_None with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=2, comp 1:

$$h_1 0^{00} = \begin{pmatrix} u_{22}^{22} & 0 & 0 \\ 0 & 2u_{22}^{12} - u_{22}^{22} & 0 \\ 0 & 0 & 2u_{22}^{02} - u_{22}^{22} \end{pmatrix}$$

$$h_1 0^{01} = \begin{pmatrix} u_{22}^{02} & 1 & 0 \\ 1 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{12} - u_{22}^{22} \end{pmatrix}$$

$$h_1 0^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 1 \\ 0 & u_{22}^{02} + u_{22}^{12} - u_{22}^{22} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_1 0^{10} = \begin{pmatrix} u_{22}^{02} & 1 & 0 \\ 1 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{12} - u_{22}^{22} \end{pmatrix}$$

$$h_1 0^{11} = \begin{pmatrix} 2u_{22}^{02} - u_{22}^{22} & 0 & 0 \\ 0 & u_{22}^{22} & 0 \\ 0 & 0 & 2u_{22}^{12} - u_{22}^{22} \end{pmatrix}$$

$$h_1 0^{12} = \begin{pmatrix} u_{22}^{02} + u_{22}^{12} - u_{22}^{22} & 0 & 0 \\ 0 & u_{22}^{02} & 1 \\ 0 & 1 & u_{22}^{12} \end{pmatrix}$$

$$h_1 0^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 1 \\ 0 & u_{22}^{02} + u_{22}^{12} - u_{22}^{22} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_1 0^{21} = \begin{pmatrix} u_{22}^{02} + u_{22}^{12} - u_{22}^{22} & 0 & 0 \\ 0 & u_{22}^{02} & 1 \\ 0 & 1 & u_{22}^{12} \end{pmatrix}$$

$$h_1 0^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} & 0 & 0 \\ 0 & 2u_{22}^{02} - u_{22}^{22} & 0 \\ 0 & 0 & u_{22}^{22} \end{pmatrix}$$

Genuine Solution no. 11 for family Z3_RR_None with pattern S=[(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)] chart=1, comp 0:

$$h_1 1^{00} = \begin{pmatrix} u_{22}^{22} & 1 & 0 \\ 1 & -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 1^{01} = \begin{pmatrix} -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} & 0 & -1 \\ 0 & u_{22}^{12} & 0 \\ -1 & 0 & u_{22}^{22} \end{pmatrix}$$

$$h_1 1^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} & -1 \\ 0 & -1 & -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} \end{pmatrix}$$

$$h_1 1^{10} = \begin{pmatrix} -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} & 0 & -1 \\ 0 & u_{22}^{12} & 0 \\ -1 & 0 & u_{22}^{22} \end{pmatrix}$$

$$h_1 1^{11} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} & 1 \\ 0 & 1 & -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} \end{pmatrix}$$

$$h_1 1^{12} = \begin{pmatrix} u_{22}^{22} & -1 & 0 \\ -1 & -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 1^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} & -1 \\ 0 & -1 & -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} \end{pmatrix}$$

$$h_1 1^{21} = \begin{pmatrix} u_{22}^{22} & -1 & 0 \\ -1 & -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 1^{22} = \begin{pmatrix} -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} & 0 & 1 \\ 0 & u_{22}^{12} & 0 \\ 1 & 0 & u_{22}^{22} \end{pmatrix}$$

Genuine Solution no. 12 for family Z3_RR_None with pattern S=[(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)] chart=1, comp 1:

$$h_1 2^{00} = \begin{pmatrix} u_{22}^{22} & 1 & 0 \\ 1 & -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 2^{01} = \begin{pmatrix} -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} & 0 & 1 \\ 0 & u_{22}^{12} & 0 \\ 1 & 0 & u_{22}^{22} \end{pmatrix}$$

$$h_1 2^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} & 1 \\ 0 & 1 & -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} \end{pmatrix}$$

$$h_1 2^{10} = \begin{pmatrix} -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} & 0 & 1 \\ 0 & u_{22}^{12} & 0 \\ 1 & 0 & u_{22}^{22} \end{pmatrix}$$

$$h_1 2^{11} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} & 1 \\ 0 & 1 & -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} \end{pmatrix}$$

$$h_1 2^{12} = \begin{pmatrix} u_{22}^{22} & 1 & 0 \\ 1 & -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 2^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} & 1 \\ 0 & 1 & -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} \end{pmatrix}$$

$$h_1 2^{21} = \begin{pmatrix} u_{22}^{22} & 1 & 0 \\ 1 & -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 2^{22} = \begin{pmatrix} -\frac{u_{22}^{12}}{4} + \frac{5u_{22}^{22}}{4} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 - 8}}{4} & 0 & 1 \\ 0 & u_{22}^{12} & 0 \\ 1 & 0 & u_{22}^{22} \end{pmatrix}$$

Genuine Solution no. 13 for family Z3_RR_None with pattern S=[(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)] chart=2, comp 0:

$$h_1 3^{00} = \begin{pmatrix} u_{22}^{22} & 0 & 1 \\ 0 & \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} & 0 \\ 1 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 3^{01} = \begin{pmatrix} \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} & 0 & 0 \\ 0 & u_{22}^{12} & -1 \\ 0 & -1 & u_{22}^{22} \end{pmatrix}$$

$$h_1 3^{02} = \begin{pmatrix} u_{22}^{12} & -1 & 0 \\ -1 & u_{22}^{22} & 0 \\ 0 & 0 & \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} \end{pmatrix}$$

$$h_1 3^{10} = \begin{pmatrix} \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} & 0 & 0 \\ 0 & u_{22}^{12} & -1 \\ 0 & -1 & u_{22}^{22} \end{pmatrix}$$

$$h_1 3^{11} = \begin{pmatrix} u_{22}^{12} & 1 & 0 \\ 1 & u_{22}^{22} & 0 \\ 0 & 0 & \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} \end{pmatrix}$$

$$h_1 3^{12} = \begin{pmatrix} u_{22}^{22} & 0 & -1 \\ 0 & \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} & 0 \\ -1 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 3^{20} = \begin{pmatrix} u_{22}^{12} & -1 & 0 \\ -1 & u_{22}^{22} & 0 \\ 0 & 0 & \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} \end{pmatrix}$$

$$h_1 3^{21} = \begin{pmatrix} u_{22}^{22} & 0 & -1 \\ 0 & \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} & 0 \\ -1 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 3^{22} = \begin{pmatrix} \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} & 0 & 0 \\ 0 & u_{22}^{12} & 1 \\ 0 & 1 & u_{22}^{22} \end{pmatrix}$$

Genuine Solution no. 14 for family Z3_RR_None with pattern S=[(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)] chart=2, comp 1:

$$h_1 4^{00} = \begin{pmatrix} u_{22}^{22} & 0 & 1 \\ 0 & \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} & 0 \\ 1 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 4^{01} = \begin{pmatrix} \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} & 0 & 0 \\ 0 & u_{22}^{12} & 1 \\ 0 & 1 & u_{22}^{22} \end{pmatrix}$$

$$h_1 4^{02} = \begin{pmatrix} u_{22}^{12} & 1 & 0 \\ 1 & u_{22}^{22} & 0 \\ 0 & 0 & \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} \end{pmatrix}$$

$$h_1 4^{10} = \begin{pmatrix} \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} & 0 & 0 \\ 0 & u_{22}^{12} & 1 \\ 0 & 1 & u_{22}^{22} \end{pmatrix}$$

$$h_1 4^{11} = \begin{pmatrix} u_{22}^{12} & 1 & 0 \\ 1 & u_{22}^{22} & 0 \\ 0 & 0 & \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} \end{pmatrix}$$

$$h_1 4^{12} = \begin{pmatrix} u_{22}^{22} & 0 & 1 \\ 0 & \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} & 0 \\ 1 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 4^{20} = \begin{pmatrix} u_{22}^{12} & 1 & 0 \\ 1 & u_{22}^{22} & 0 \\ 0 & 0 & \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} \end{pmatrix}$$

$$h_1 4^{21} = \begin{pmatrix} u_{22}^{22} & 0 & 1 \\ 0 & \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} & 0 \\ 1 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 4^{22} = \begin{pmatrix} \frac{u_{22}^{12}}{2} + \frac{u_{22}^{22}}{2} - \frac{\sqrt{9\left(u_{22}^{12}\right)^2 - 18u_{22}^{12}u_{22}^{22} + 9\left(u_{22}^{22}\right)^2 + 4}}{2} & 0 & 0 \\ 0 & u_{22}^{12} & 1 \\ 0 & 1 & u_{22}^{22} \end{pmatrix}$$

2 Z3_RR diagonális mértékkel

Genuine Solution no. 0 for family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 0:

$$h_0^{00} = \begin{pmatrix} u_{22}^{22} & \left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}+1\right)\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & -\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} \\ \left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}+1\right)\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & u_{22}^{11} & 1 \\ -\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & 1 & -\frac{2u_{22}^{11}u_{22}^{12}-3u_{22}^{11}u_{22}^{22}+3u_{22}^{11}-4\left(u_{22}^{12}\right)^2+8u_{22}^{12}u_{22}^{22}-8u_{22}^{12}-3\left(u_{22}^{22}\right)^2+6u_{22}^{22}-4}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1} \end{pmatrix}$$

$$h_0^{01} = \begin{pmatrix} -u_{22}^{12}+2u_{22}^{22}-2 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22}-1 \end{pmatrix}$$

$$h_0^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22}-1 & 0 \\ 0 & 0 & -u_{22}^{12}+2u_{22}^{22}-2 \end{pmatrix}$$

$$h_0^{10} = \begin{pmatrix} -u_{22}^{12}+2u_{22}^{22}-2 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22}-1 \end{pmatrix}$$

$$h_0^{11} = \begin{pmatrix} -\frac{2u_{22}^{11}u_{22}^{12}-3u_{22}^{11}u_{22}^{22}+3u_{22}^{11}-4\left(u_{22}^{12}\right)^2+8u_{22}^{12}u_{22}^{22}-8u_{22}^{12}-3\left(u_{22}^{22}\right)^2+6u_{22}^{22}-4}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1} & -\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & 1 \\ -\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & u_{22}^{22} & \left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}+1\right)\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} \\ 1 & \left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}+1\right)\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & u_{22}^{11} \end{pmatrix}$$

$$h_0^{12} = \begin{pmatrix} u_{22}^{22}-1 & 0 & 0 \\ 0 & -u_{22}^{12}+2u_{22}^{22}-2 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_0^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22}-1 & 0 \\ 0 & 0 & -u_{22}^{12}+2u_{22}^{22}-2 \end{pmatrix}$$

$$h_0^{21} = \begin{pmatrix} u_{22}^{22}-1 & 0 & 0 \\ 0 & -u_{22}^{12}+2u_{22}^{22}-2 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_0^{22} = \begin{pmatrix} u_{22}^{11} & 1 & \left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}+1\right)\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} \\ 1 & -\frac{2u_{22}^{11}u_{22}^{12}-3u_{22}^{11}u_{22}^{22}+3u_{22}^{11}-4\left(u_{22}^{12}\right)^2+8u_{22}^{12}u_{22}^{22}-8u_{22}^{12}-3\left(u_{22}^{22}\right)^2+6u_{22}^{22}-4}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1} & -\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} \\ \left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}+1\right)\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & -\sqrt{\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}-1}} & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 0 in family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 0:

$$
\begin{aligned}
\gamma_0^{(0,0)} &= g_{22} \\
\gamma_0^{(0,1)} &= -g_{10} + 2g_{22} \\
\gamma_0^{(0,2)} &= -g_{20} + 2g_{22} \\
\gamma_0^{(1,0)} &= g_{10} \\
\gamma_0^{(1,1)} &= g_{22} \\
\gamma_0^{(1,2)} &= -g_{21} + 2g_{22} \\
\gamma_0^{(2,0)} &= g_{20} \\
\gamma_0^{(2,1)} &= g_{21} \\
\gamma_0^{(2,2)} &= g_{22}
\end{aligned}
$$

Genuine Solution no. 1 for family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 1:

$$h_1^{00} = \begin{pmatrix} u_{22}^{22} & \sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}}\left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}-1\right) & -\sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}} \\ \sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}}\left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}-1\right) & u_{22}^{11} & 1 \\ -\sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}} & 1 & -\frac{2u_{22}^{11}u_{22}^{12}-3u_{22}^{11}u_{22}^{22}-3u_{22}^{11}-4\left(u_{22}^{12}\right)^2+8u_{22}^{12}u_{22}^{22}+8u_{22}^{12}-3\left(u_{22}^{22}\right)^2-6u_{22}^{22}-4}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1} \end{pmatrix}$$

$$h_1^{01} = \begin{pmatrix} -u_{22}^{12}+2u_{22}^{22}+2 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22}+1 \end{pmatrix}$$

$$h_1^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22}+1 & 0 \\ 0 & 0 & -u_{22}^{12}+2u_{22}^{22}+2 \end{pmatrix}$$

$$h_1^{10} = \begin{pmatrix} -u_{22}^{12}+2u_{22}^{22}+2 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22}+1 \end{pmatrix}$$

$$h_1^{11} = \begin{pmatrix} -\frac{2u_{22}^{11}u_{22}^{12}-3u_{22}^{11}u_{22}^{22}-3u_{22}^{11}-4\left(u_{22}^{12}\right)^2+8u_{22}^{12}u_{22}^{22}+8u_{22}^{12}-3\left(u_{22}^{22}\right)^2-6u_{22}^{22}-4}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1} & -\sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}} & 1 \\ -\sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}} & u_{22}^{22} & \sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}}\left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}-1\right) \\ 1 & \sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}}\left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}-1\right) & u_{22}^{11} \end{pmatrix}$$

$$h_1^{12} = \begin{pmatrix} u_{22}^{22}+1 & 0 & 0 \\ 0 & -u_{22}^{12}+2u_{22}^{22}+2 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22}+1 & 0 \\ 0 & 0 & -u_{22}^{12}+2u_{22}^{22}+2 \end{pmatrix}$$

$$h_1^{21} = \begin{pmatrix} u_{22}^{22}+1 & 0 & 0 \\ 0 & -u_{22}^{12}+2u_{22}^{22}+2 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1^{22} = \begin{pmatrix} u_{22}^{11} & 1 & \sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}}\left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}-1\right) \\ 1 & -\frac{2u_{22}^{11}u_{22}^{12}-3u_{22}^{11}u_{22}^{22}-3u_{22}^{11}-4\left(u_{22}^{12}\right)^2+8u_{22}^{12}u_{22}^{22}+8u_{22}^{12}-3\left(u_{22}^{22}\right)^2-6u_{22}^{22}-4}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1} & -\sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}} \\ \sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}}\left(-u_{22}^{11}+2u_{22}^{12}-u_{22}^{22}-1\right) & -\sqrt{-\frac{1}{u_{22}^{11}-2u_{22}^{12}+u_{22}^{22}+1}} & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 1 in family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 1:

$$
\begin{aligned}
\gamma_1^{(0,0)} &= g_{22} \\
\gamma_1^{(0,1)} &= -g_{10} + 2g_{22} \\
\gamma_1^{(0,2)} &= -g_{20} + 2g_{22} \\
\gamma_1^{(1,0)} &= g_{10} \\
\gamma_1^{(1,1)} &= g_{22} \\
\gamma_1^{(1,2)} &= -g_{21} + 2g_{22} \\
\gamma_1^{(2,0)} &= g_{20} \\
\gamma_1^{(2,1)} &= g_{21} \\
\gamma_1^{(2,2)} &= g_{22}
\end{aligned}
$$

Genuine Solution no. 3 for family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 3:

$$h_3^{00} = \begin{pmatrix} u_{22}^{22} & 1 & 1 \\ 1 & 2u_{22}^{12} - u_{22}^{22} + 4 & 1 \\ 1 & 1 & -2u_{22}^{12} + 3u_{22}^{22} - 4 \end{pmatrix}$$

$$h_3^{01} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} - 2 \end{pmatrix}$$

$$h_3^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} - 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 \end{pmatrix}$$

$$h_3^{10} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} - 2 \end{pmatrix}$$

$$h_3^{11} = \begin{pmatrix} -2u_{22}^{12} + 3u_{22}^{22} - 4 & 1 & 1 \\ 1 & u_{22}^{22} & 1 \\ 1 & 1 & 2u_{22}^{12} - u_{22}^{22} + 4 \end{pmatrix}$$

$$h_3^{12} = \begin{pmatrix} u_{22}^{22} - 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_3^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} - 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 \end{pmatrix}$$

$$h_3^{21} = \begin{pmatrix} u_{22}^{22} - 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_3^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} + 4 & 1 & 1 \\ 1 & -2u_{22}^{12} + 3u_{22}^{22} - 4 & 1 \\ 1 & 1 & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 3 in family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 3:

$$
\begin{aligned}
\gamma_3^{(0,0)} &= g_{22} \\
\gamma_3^{(0,1)} &= -g_{10} + 2g_{22} \\
\gamma_3^{(0,2)} &= -g_{20} + 2g_{22} \\
\gamma_3^{(1,0)} &= g_{10} \\
\gamma_3^{(1,1)} &= g_{22} \\
\gamma_3^{(1,2)} &= -g_{21} + 2g_{22} \\
\gamma_3^{(2,0)} &= g_{20} \\
\gamma_3^{(2,1)} &= g_{21} \\
\gamma_3^{(2,2)} &= g_{22}
\end{aligned}
$$

Genuine Solution no. 4 for family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 4:

$$h_4^{00} = \begin{pmatrix} u_{22}^{22} & -1 & 1 \\ -1 & 2u_{22}^{12} - u_{22}^{22} - 4 & 1 \\ 1 & 1 & -2u_{22}^{12} + 3u_{22}^{22} + 4 \end{pmatrix}$$

$$h_4^{01} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} + 2 \end{pmatrix}$$

$$h_4^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} + 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 \end{pmatrix}$$

$$h_4^{10} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} + 2 \end{pmatrix}$$

$$h_4^{11} = \begin{pmatrix} -2u_{22}^{12} + 3u_{22}^{22} + 4 & 1 & 1 \\ 1 & u_{22}^{22} & -1 \\ 1 & -1 & 2u_{22}^{12} - u_{22}^{22} - 4 \end{pmatrix}$$

$$h_4^{12} = \begin{pmatrix} u_{22}^{22} + 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_4^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} + 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 \end{pmatrix}$$

$$h_4^{21} = \begin{pmatrix} u_{22}^{22} + 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_4^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} - 4 & 1 & -1 \\ 1 & -2u_{22}^{12} + 3u_{22}^{22} + 4 & 1 \\ -1 & 1 & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 4 in family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 4:

$$
\begin{aligned}
\gamma_4^{(0,0)} &= g_{22} \\
\gamma_4^{(0,1)} &= -g_{10} + 2g_{22} \\
\gamma_4^{(0,2)} &= -g_{20} + 2g_{22} \\
\gamma_4^{(1,0)} &= g_{10} \\
\gamma_4^{(1,1)} &= g_{22} \\
\gamma_4^{(1,2)} &= -g_{21} + 2g_{22} \\
\gamma_4^{(2,0)} &= g_{20} \\
\gamma_4^{(2,1)} &= g_{21} \\
\gamma_4^{(2,2)} &= g_{22}
\end{aligned}
$$

Genuine Solution no. 5 for family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 5:

$$h_5^{00} = \begin{pmatrix} u_{22}^{22} & -\frac{\sqrt{2+2\sqrt{5}}}{2} & -\sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} \\ -\frac{\sqrt{2+2\sqrt{5}}}{2} & 2u_{22}^{12} - u_{22}^{22} + 2 + \sqrt{5} & 1 \\ -\sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & 1 & -2u_{22}^{12} + 3u_{22}^{22} - \frac{3\sqrt{5}}{2} - \frac{1}{2} \end{pmatrix}$$

$$h_5^{01} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} - \frac{5\sqrt{5}}{4} - \frac{5}{4} & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} - \frac{3\sqrt{5}}{4} - \frac{3}{4} \end{pmatrix}$$

$$h_5^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} - \frac{3\sqrt{5}}{4} - \frac{3}{4} & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} - \frac{5\sqrt{5}}{4} - \frac{5}{4} \end{pmatrix}$$

$$h_5^{10} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} - \frac{5\sqrt{5}}{4} - \frac{5}{4} & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} - \frac{3\sqrt{5}}{4} - \frac{3}{4} \end{pmatrix}$$

$$h_5^{11} = \begin{pmatrix} -2u_{22}^{12} + 3u_{22}^{22} - \frac{3\sqrt{5}}{2} - \frac{1}{2} & -\sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & 1 \\ -\sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & u_{22}^{22} & -\frac{\sqrt{2+2\sqrt{5}}}{2} \\ 1 & -\frac{\sqrt{2+2\sqrt{5}}}{2} & 2u_{22}^{12} - u_{22}^{22} + 2 + \sqrt{5} \end{pmatrix}$$

$$h_5^{12} = \begin{pmatrix} u_{22}^{22} - \frac{3\sqrt{5}}{4} - \frac{3}{4} & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} - \frac{5\sqrt{5}}{4} - \frac{5}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_5^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} - \frac{3\sqrt{5}}{4} - \frac{3}{4} & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} - \frac{5\sqrt{5}}{4} - \frac{5}{4} \end{pmatrix}$$

$$h_5^{21} = \begin{pmatrix} u_{22}^{22} - \frac{3\sqrt{5}}{4} - \frac{3}{4} & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} - \frac{5\sqrt{5}}{4} - \frac{5}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_5^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} + 2 + \sqrt{5} & 1 & -\frac{\sqrt{2+2\sqrt{5}}}{2} \\ 1 & -2u_{22}^{12} + 3u_{22}^{22} - \frac{3\sqrt{5}}{2} - \frac{1}{2} & -\sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} \\ -\frac{\sqrt{2+2\sqrt{5}}}{2} & -\sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 5 in family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 5:

$$
\begin{aligned}
\gamma_5^{(0,0)} &= g_{22} \\
\gamma_5^{(0,1)} &= -g_{10} + 2g_{22} \\
\gamma_5^{(0,2)} &= -g_{20} + 2g_{22} \\
\gamma_5^{(1,0)} &= g_{10} \\
\gamma_5^{(1,1)} &= g_{22} \\
\gamma_5^{(1,2)} &= -g_{21} + 2g_{22} \\
\gamma_5^{(2,0)} &= g_{20} \\
\gamma_5^{(2,1)} &= g_{21} \\
\gamma_5^{(2,2)} &= g_{22}
\end{aligned}
$$

Genuine Solution no. 6 for family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 6:

$$h_6^{00} = \begin{pmatrix} u_{22}^{22} & -\frac{\sqrt{2+2\sqrt{5}}}{2} & \sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} \\ -\frac{\sqrt{2+2\sqrt{5}}}{2} & 2u_{22}^{12} - u_{22}^{22} - \sqrt{5} - 2 & 1 \\ \sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & 1 & -2u_{22}^{12} + 3u_{22}^{22} + \frac{1}{2} + \frac{3\sqrt{5}}{2} \end{pmatrix}$$

$$h_6^{01} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} + \frac{5}{4} + \frac{5\sqrt{5}}{4} & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} + \frac{3}{4} + \frac{3\sqrt{5}}{4} \end{pmatrix}$$

$$h_6^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} + \frac{3}{4} + \frac{3\sqrt{5}}{4} & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} + \frac{5}{4} + \frac{5\sqrt{5}}{4} \end{pmatrix}$$

$$h_6^{10} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} + \frac{5}{4} + \frac{5\sqrt{5}}{4} & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} + \frac{3}{4} + \frac{3\sqrt{5}}{4} \end{pmatrix}$$

$$h_6^{11} = \begin{pmatrix} -2u_{22}^{12} + 3u_{22}^{22} + \frac{1}{2} + \frac{3\sqrt{5}}{2} & \sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & 1 \\ \sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & u_{22}^{22} & -\frac{\sqrt{2+2\sqrt{5}}}{2} \\ 1 & -\frac{\sqrt{2+2\sqrt{5}}}{2} & 2u_{22}^{12} - u_{22}^{22} - \sqrt{5} - 2 \end{pmatrix}$$

$$h_6^{12} = \begin{pmatrix} u_{22}^{22} + \frac{3}{4} + \frac{3\sqrt{5}}{4} & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} + \frac{5}{4} + \frac{5\sqrt{5}}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_6^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} + \frac{3}{4} + \frac{3\sqrt{5}}{4} & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} + \frac{5}{4} + \frac{5\sqrt{5}}{4} \end{pmatrix}$$

$$h_6^{21} = \begin{pmatrix} u_{22}^{22} + \frac{3}{4} + \frac{3\sqrt{5}}{4} & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} + \frac{5}{4} + \frac{5\sqrt{5}}{4} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_6^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} - \sqrt{5} - 2 & 1 & -\frac{\sqrt{2+2\sqrt{5}}}{2} \\ 1 & -2u_{22}^{12} + 3u_{22}^{22} + \frac{1}{2} + \frac{3\sqrt{5}}{2} & \sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} \\ -\frac{\sqrt{2+2\sqrt{5}}}{2} & \sqrt{\frac{1}{2}+\frac{\sqrt{5}}{2}} & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 6 in family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 6:

$$
\begin{aligned}
\gamma_6^{(0,0)} &= g_{22} \\
\gamma_6^{(0,1)} &= -g_{10} + 2g_{22} \\
\gamma_6^{(0,2)} &= -g_{20} + 2g_{22} \\
\gamma_6^{(1,0)} &= g_{10} \\
\gamma_6^{(1,1)} &= g_{22} \\
\gamma_6^{(1,2)} &= -g_{21} + 2g_{22} \\
\gamma_6^{(2,0)} &= g_{20} \\
\gamma_6^{(2,1)} &= g_{21} \\
\gamma_6^{(2,2)} &= g_{22}
\end{aligned}
$$

Genuine Solution no. 7 for family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 7:

$$h_7^{00} = \begin{pmatrix} u_{22}^{22} & -1 & -1 \\ -1 & 2u_{22}^{12} - u_{22}^{22} + 4 & 1 \\ -1 & 1 & -2u_{22}^{12} + 3u_{22}^{22} - 4 \end{pmatrix}$$

$$h_7^{01} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} - 2 \end{pmatrix}$$

$$h_7^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} - 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 \end{pmatrix}$$

$$h_7^{10} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} - 2 \end{pmatrix}$$

$$h_7^{11} = \begin{pmatrix} -2u_{22}^{12} + 3u_{22}^{22} - 4 & -1 & 1 \\ -1 & u_{22}^{22} & -1 \\ 1 & -1 & 2u_{22}^{12} - u_{22}^{22} + 4 \end{pmatrix}$$

$$h_7^{12} = \begin{pmatrix} u_{22}^{22} - 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_7^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} - 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 \end{pmatrix}$$

$$h_7^{21} = \begin{pmatrix} u_{22}^{22} - 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} - 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_7^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} + 4 & 1 & -1 \\ 1 & -2u_{22}^{12} + 3u_{22}^{22} - 4 & -1 \\ -1 & -1 & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 7 in family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 7:

$$
\begin{aligned}
\gamma_7^{(0,0)} &= g_{22} \\
\gamma_7^{(0,1)} &= -g_{10} + 2g_{22} \\
\gamma_7^{(0,2)} &= -g_{20} + 2g_{22} \\
\gamma_7^{(1,0)} &= g_{10} \\
\gamma_7^{(1,1)} &= g_{22} \\
\gamma_7^{(1,2)} &= -g_{21} + 2g_{22} \\
\gamma_7^{(2,0)} &= g_{20} \\
\gamma_7^{(2,1)} &= g_{21} \\
\gamma_7^{(2,2)} &= g_{22}
\end{aligned}
$$

Genuine Solution no. 8 for family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 8:

$$h_8^{00} = \begin{pmatrix} u_{22}^{22} & 1 & -1 \\ 1 & 2u_{22}^{12} - u_{22}^{22} - 4 & 1 \\ -1 & 1 & -2u_{22}^{12} + 3u_{22}^{22} + 4 \end{pmatrix}$$

$$h_8^{01} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} + 2 \end{pmatrix}$$

$$h_8^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} + 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 \end{pmatrix}$$

$$h_8^{10} = \begin{pmatrix} -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 & 0 \\ 0 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} + 2 \end{pmatrix}$$

$$h_8^{11} = \begin{pmatrix} -2u_{22}^{12} + 3u_{22}^{22} + 4 & -1 & 1 \\ -1 & u_{22}^{22} & 1 \\ 1 & 1 & 2u_{22}^{12} - u_{22}^{22} - 4 \end{pmatrix}$$

$$h_8^{12} = \begin{pmatrix} u_{22}^{22} + 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_8^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} + 2 & 0 \\ 0 & 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 \end{pmatrix}$$

$$h_8^{21} = \begin{pmatrix} u_{22}^{22} + 2 & 0 & 0 \\ 0 & -u_{22}^{12} + 2u_{22}^{22} + 4 & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_8^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} - 4 & 1 & 1 \\ 1 & -2u_{22}^{12} + 3u_{22}^{22} + 4 & -1 \\ 1 & -1 & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 8 in family Z3_RR_diag-gauge with pattern S=[(0, 0), (1, 1), (2, 2)] chart=0, comp 8:

$$
\begin{aligned}
\gamma_8^{(0,0)} &= g_{22} \\
\gamma_8^{(0,1)} &= -g_{10} + 2g_{22} \\
\gamma_8^{(0,2)} &= -g_{20} + 2g_{22} \\
\gamma_8^{(1,0)} &= g_{10} \\
\gamma_8^{(1,1)} &= g_{22} \\
\gamma_8^{(1,2)} &= -g_{21} + 2g_{22} \\
\gamma_8^{(2,0)} &= g_{20} \\
\gamma_8^{(2,1)} &= g_{21} \\
\gamma_8^{(2,2)} &= g_{22}
\end{aligned}
$$

Genuine Solution no. 9 for family Z3_RR_diag-gauge with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=2, comp 0:

$$h_9^{00} = \begin{pmatrix} u_{22}^{22} & 0 & 0 \\ 0 & u_{22}^{11} & 0 \\ 0 & 0 & -u_{22}^{11} + 2u_{22}^{22} \end{pmatrix}$$

$$h_9^{01} = \begin{pmatrix} -u_{22}^{11} + u_{22}^{12} + u_{22}^{22} & 1 & 0 \\ 1 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} \end{pmatrix}$$

$$h_9^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 1 \\ 0 & u_{22}^{22} & 0 \\ 1 & 0 & -u_{22}^{11} + u_{22}^{12} + u_{22}^{22} \end{pmatrix}$$

$$h_9^{10} = \begin{pmatrix} -u_{22}^{11} + u_{22}^{12} + u_{22}^{22} & 1 & 0 \\ 1 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{22} \end{pmatrix}$$

$$h_9^{11} = \begin{pmatrix} -u_{22}^{11} + 2u_{22}^{22} & 0 & 0 \\ 0 & u_{22}^{22} & 0 \\ 0 & 0 & u_{22}^{11} \end{pmatrix}$$

$$h_9^{12} = \begin{pmatrix} u_{22}^{22} & 0 & 0 \\ 0 & -u_{22}^{11} + u_{22}^{12} + u_{22}^{22} & 1 \\ 0 & 1 & u_{22}^{12} \end{pmatrix}$$

$$h_9^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 1 \\ 0 & u_{22}^{22} & 0 \\ 1 & 0 & -u_{22}^{11} + u_{22}^{12} + u_{22}^{22} \end{pmatrix}$$

$$h_9^{21} = \begin{pmatrix} u_{22}^{22} & 0 & 0 \\ 0 & -u_{22}^{11} + u_{22}^{12} + u_{22}^{22} & 1 \\ 0 & 1 & u_{22}^{12} \end{pmatrix}$$

$$h_9^{22} = \begin{pmatrix} u_{22}^{11} & 0 & 0 \\ 0 & -u_{22}^{11} + 2u_{22}^{22} & 0 \\ 0 & 0 & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 9 in family Z3_RR_diag-gauge with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=2, comp 0:

$$
\begin{aligned}
\gamma_9^{(0,0)} &= g_{22} \\
\gamma_9^{(0,1)} &= g_{01} \\
\gamma_9^{(0,2)} &= g_{02} \\
\gamma_9^{(1,0)} &= g_{10} \\
\gamma_9^{(1,1)} &= g_{22} \\
\gamma_9^{(1,2)} &= g_{12} \\
\gamma_9^{(2,0)} &= g_{20} \\
\gamma_9^{(2,1)} &= g_{21} \\
\gamma_9^{(2,2)} &= g_{22}
\end{aligned}
$$

Genuine Solution no. 10 for family Z3_RR_diag-gauge with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=2, comp 1:

$$h_1 0^{00} = \begin{pmatrix} u_{22}^{22} & 0 & 0 \\ 0 & 2u_{22}^{12} - u_{22}^{22} & 0 \\ 0 & 0 & 2u_{22}^{02} - u_{22}^{22} \end{pmatrix}$$

$$h_1 0^{01} = \begin{pmatrix} u_{22}^{02} & 1 & 0 \\ 1 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{12} - u_{22}^{22} \end{pmatrix}$$

$$h_1 0^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 1 \\ 0 & u_{22}^{02} + u_{22}^{12} - u_{22}^{22} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_1 0^{10} = \begin{pmatrix} u_{22}^{02} & 1 & 0 \\ 1 & u_{22}^{12} & 0 \\ 0 & 0 & u_{22}^{02} + u_{22}^{12} - u_{22}^{22} \end{pmatrix}$$

$$h_1 0^{11} = \begin{pmatrix} 2u_{22}^{02} - u_{22}^{22} & 0 & 0 \\ 0 & u_{22}^{22} & 0 \\ 0 & 0 & 2u_{22}^{12} - u_{22}^{22} \end{pmatrix}$$

$$h_1 0^{12} = \begin{pmatrix} u_{22}^{02} + u_{22}^{12} - u_{22}^{22} & 0 & 0 \\ 0 & u_{22}^{02} & 1 \\ 0 & 1 & u_{22}^{12} \end{pmatrix}$$

$$h_1 0^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 1 \\ 0 & u_{22}^{02} + u_{22}^{12} - u_{22}^{22} & 0 \\ 1 & 0 & u_{22}^{02} \end{pmatrix}$$

$$h_1 0^{21} = \begin{pmatrix} u_{22}^{02} + u_{22}^{12} - u_{22}^{22} & 0 & 0 \\ 0 & u_{22}^{02} & 1 \\ 0 & 1 & u_{22}^{12} \end{pmatrix}$$

$$h_1 0^{22} = \begin{pmatrix} 2u_{22}^{12} - u_{22}^{22} & 0 & 0 \\ 0 & 2u_{22}^{02} - u_{22}^{22} & 0 \\ 0 & 0 & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 10 in family Z3_RR_diag-gauge with pattern S=[(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)] chart=2, comp 1:

$$
\begin{aligned}
\gamma_1^{(0,0)}0 &= g_{22} \\
\gamma_1^{(0,1)}0 &= g_{01} \\
\gamma_1^{(0,2)}0 &= g_{02} \\
\gamma_1^{(1,0)}0 &= g_{10} \\
\gamma_1^{(1,1)}0 &= g_{22} \\
\gamma_1^{(1,2)}0 &= g_{12} \\
\gamma_1^{(2,0)}0 &= g_{20} \\
\gamma_1^{(2,1)}0 &= g_{21} \\
\gamma_1^{(2,2)}0 &= g_{22}
\end{aligned}
$$

Genuine Solution no. 11 for family Z3_RR_diag-gauge with pattern S=[(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)] chart=1, comp 0:

$$h_1 1^{00} = \begin{pmatrix} u_{22}^{22} & 1 & 0 \\ 1 & u_{22}^{11} & 0 \\ 0 & 0 & -\frac{3u_{22}^{11}}{2} + u_{22}^{12} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11}\right)^2 - 4u_{22}^{11}u_{22}^{12} + 2u_{22}^{11}u_{22}^{22} + 4\left(u_{22}^{12}\right)^2 - 4u_{22}^{12}u_{22}^{22} + \left(u_{22}^{22}\right)^2 - 4}}{2} \end{pmatrix}$$

$$h_1 1^{01} = \begin{pmatrix} -\frac{u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} - 2\right)\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} + 2\right)}}{2} & 0 & -1 \\ 0 & u_{22}^{12} & 0 \\ -1 & 0 & u_{22}^{22} \end{pmatrix}$$

$$h_1 1^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} & -1 \\ 0 & -1 & -\frac{u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} - 2\right)\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} + 2\right)}}{2} \end{pmatrix}$$

$$h_1 1^{10} = \begin{pmatrix} -\frac{u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} - 2\right)\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} + 2\right)}}{2} & 0 & -1 \\ 0 & u_{22}^{12} & 0 \\ -1 & 0 & u_{22}^{22} \end{pmatrix}$$

$$h_1 1^{11} = \begin{pmatrix} -\frac{3u_{22}^{11}}{2} + u_{22}^{12} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11}\right)^2 - 4u_{22}^{11}u_{22}^{12} + 2u_{22}^{11}u_{22}^{22} + 4\left(u_{22}^{12}\right)^2 - 4u_{22}^{12}u_{22}^{22} + \left(u_{22}^{22}\right)^2 - 4}}{2} & 0 & 0 \\ 0 & u_{22}^{22} & 1 \\ 0 & 1 & u_{22}^{11} \end{pmatrix}$$

$$h_1 1^{12} = \begin{pmatrix} u_{22}^{22} & -1 & 0 \\ -1 & -\frac{u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} - 2\right)\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} + 2\right)}}{2} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 1^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} & -1 \\ 0 & -1 & -\frac{u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} - 2\right)\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} + 2\right)}}{2} \end{pmatrix}$$

$$h_1 1^{21} = \begin{pmatrix} u_{22}^{22} & -1 & 0 \\ -1 & -\frac{u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} - 2\right)\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} + 2\right)}}{2} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 1^{22} = \begin{pmatrix} u_{22}^{11} & 0 & 1 \\ 0 & -\frac{3u_{22}^{11}}{2} + u_{22}^{12} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11}\right)^2 - 4u_{22}^{11}u_{22}^{12} + 2u_{22}^{11}u_{22}^{22} + 4\left(u_{22}^{12}\right)^2 - 4u_{22}^{12}u_{22}^{22} + \left(u_{22}^{22}\right)^2 - 4}}{2} & 0 \\ 1 & 0 & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 11 in family Z3_RR_diag-gauge with pattern S=[(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)] chart=1, comp 0:

$$\gamma_1^{(0,0)}1 = g_{22}$$

$$\gamma_1^{(0,1)}1 = -g_{20} + g_{21} + g_{22} - \frac{3u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11}\right)^2 - 4u_{22}^{11}u_{22}^{12} + 2u_{22}^{11}u_{22}^{22} + 4\left(u_{22}^{12}\right)^2 - 4u_{22}^{12}u_{22}^{22} + \left(u_{22}^{22}\right)^2 - 4}}{2}$$

$$\gamma_1^{(0,2)}1 = -g_{20} + 2g_{22}$$

$$\gamma_1^{(1,0)}1 = g_{20} - g_{21} + g_{22} + \frac{3u_{22}^{11}}{2} - \frac{3u_{22}^{22}}{2} + \frac{\sqrt{\left(u_{22}^{11}\right)^2 - 4u_{22}^{11}u_{22}^{12} + 2u_{22}^{11}u_{22}^{22} + 4\left(u_{22}^{12}\right)^2 - 4u_{22}^{12}u_{22}^{22} + \left(u_{22}^{22}\right)^2 - 4}}{2}$$

$$\gamma_1^{(1,1)}1 = g_{22}$$

$$\gamma_1^{(1,2)}1 = -g_{21} + 2g_{22}$$

$$\gamma_1^{(2,0)}1 = g_{20}$$

$$\gamma_1^{(2,1)}1 = g_{21}$$

$$\gamma_1^{(2,2)}1 = g_{22}$$

Genuine Solution no. 12 for family Z3_RR_diag-gauge with pattern S=[(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)] chart=1, comp 1:

$$h_1 2^{00} = \begin{pmatrix} u_{22}^{22} & 1 & 0 \\ 1 & u_{22}^{11} & 0 \\ 0 & 0 & -\frac{3u_{22}^{11}}{2} + u_{22}^{12} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11}\right)^2 - 4u_{22}^{11}u_{22}^{12} + 2u_{22}^{11}u_{22}^{22} + 4\left(u_{22}^{12}\right)^2 - 4u_{22}^{12}u_{22}^{22} + \left(u_{22}^{22}\right)^2 - 4}}{2} \end{pmatrix}$$

$$h_1 2^{01} = \begin{pmatrix} -\frac{u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} - 2\right)\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} + 2\right)}}{2} & 0 & 1 \\ 0 & u_{22}^{12} & 0 \\ 1 & 0 & u_{22}^{22} \end{pmatrix}$$

$$h_1 2^{02} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} & 1 \\ 0 & 1 & -\frac{u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} - 2\right)\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} + 2\right)}}{2} \end{pmatrix}$$

$$h_1 2^{10} = \begin{pmatrix} -\frac{u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} - 2\right)\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} + 2\right)}}{2} & 0 & 1 \\ 0 & u_{22}^{12} & 0 \\ 1 & 0 & u_{22}^{22} \end{pmatrix}$$

$$h_1 2^{11} = \begin{pmatrix} -\frac{3u_{22}^{11}}{2} + u_{22}^{12} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11}\right)^2 - 4u_{22}^{11}u_{22}^{12} + 2u_{22}^{11}u_{22}^{22} + 4\left(u_{22}^{12}\right)^2 - 4u_{22}^{12}u_{22}^{22} + \left(u_{22}^{22}\right)^2 - 4}}{2} & 0 & 0 \\ 0 & u_{22}^{22} & 1 \\ 0 & 1 & u_{22}^{11} \end{pmatrix}$$

$$h_1 2^{12} = \begin{pmatrix} u_{22}^{22} & 1 & 0 \\ 1 & -\frac{u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} - 2\right)\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} + 2\right)}}{2} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 2^{20} = \begin{pmatrix} u_{22}^{12} & 0 & 0 \\ 0 & u_{22}^{22} & 1 \\ 0 & 1 & -\frac{u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} - 2\right)\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} + 2\right)}}{2} \end{pmatrix}$$

$$h_1 2^{21} = \begin{pmatrix} u_{22}^{22} & 1 & 0 \\ 1 & -\frac{u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} - 2\right)\left(u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22} + 2\right)}}{2} & 0 \\ 0 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 2^{22} = \begin{pmatrix} u_{22}^{11} & 0 & 1 \\ 0 & -\frac{3u_{22}^{11}}{2} + u_{22}^{12} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11}\right)^2 - 4u_{22}^{11}u_{22}^{12} + 2u_{22}^{11}u_{22}^{22} + 4\left(u_{22}^{12}\right)^2 - 4u_{22}^{12}u_{22}^{22} + \left(u_{22}^{22}\right)^2 - 4}}{2} & 0 \\ 1 & 0 & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 12 in family Z3_RR_diag-gauge with pattern S=[(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)] chart=1, comp 1:

$$\gamma_1^{(0,0)}2 = g_{22}$$

$$\gamma_1^{(0,1)}2 = -g_{20} + g_{21} + g_{22} - \frac{3u_{22}^{11}}{2} + \frac{3u_{22}^{22}}{2} - \frac{\sqrt{\left(u_{22}^{11}\right)^2 - 4u_{22}^{11}u_{22}^{12} + 2u_{22}^{11}u_{22}^{22} + 4\left(u_{22}^{12}\right)^2 - 4u_{22}^{12}u_{22}^{22} + \left(u_{22}^{22}\right)^2 - 4}}{2}$$

$$\gamma_1^{(0,2)}2 = -g_{20} + 2g_{22}$$

$$\gamma_1^{(1,0)}2 = g_{20} - g_{21} + g_{22} + \frac{3u_{22}^{11}}{2} - \frac{3u_{22}^{22}}{2} + \frac{\sqrt{\left(u_{22}^{11}\right)^2 - 4u_{22}^{11}u_{22}^{12} + 2u_{22}^{11}u_{22}^{22} + 4\left(u_{22}^{12}\right)^2 - 4u_{22}^{12}u_{22}^{22} + \left(u_{22}^{22}\right)^2 - 4}}{2}$$

$$\gamma_1^{(1,1)}2 = g_{22}$$

$$\gamma_1^{(1,2)}2 = -g_{21} + 2g_{22}$$

$$\gamma_1^{(2,0)}2 = g_{20}$$

$$\gamma_1^{(2,1)}2 = g_{21}$$

$$\gamma_1^{(2,2)}2 = g_{22}$$

Genuine Solution no. 13 for family Z3_RR_diag-gauge with pattern S=[(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)] chart=2, comp 0:

$$h_1 3^{00} = \begin{pmatrix} u_{22}^{22} & 0 & 1 \\ 0 & u_{22}^{11} & 0 \\ 1 & 0 & -\frac{\left(u_{22}^{11}\right)^2 - 2u_{22}^{11}u_{22}^{12} - u_{22}^{11}u_{22}^{22} + 4u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} \end{pmatrix}$$

$$h_1 3^{01} = \begin{pmatrix} -\frac{u_{22}^{11}u_{22}^{12} - 2u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} & 0 & 0 \\ 0 & u_{22}^{12} & -1 \\ 0 & -1 & u_{22}^{22} \end{pmatrix}$$

$$h_1 3^{02} = \begin{pmatrix} u_{22}^{12} & -1 & 0 \\ -1 & u_{22}^{22} & 0 \\ 0 & 0 & -\frac{u_{22}^{11}u_{22}^{12} - 2u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} \end{pmatrix}$$

$$h_1 3^{10} = \begin{pmatrix} -\frac{u_{22}^{11}u_{22}^{12} - 2u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} & 0 & 0 \\ 0 & u_{22}^{12} & -1 \\ 0 & -1 & u_{22}^{22} \end{pmatrix}$$

$$h_1 3^{11} = \begin{pmatrix} -\frac{\left(u_{22}^{11}\right)^2 - 2u_{22}^{11}u_{22}^{12} - u_{22}^{11}u_{22}^{22} + 4u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} & 1 & 0 \\ 1 & u_{22}^{22} & 0 \\ 0 & 0 & u_{22}^{11} \end{pmatrix}$$

$$h_1 3^{12} = \begin{pmatrix} u_{22}^{22} & 0 & -1 \\ 0 & -\frac{u_{22}^{11}u_{22}^{12} - 2u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} & 0 \\ -1 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 3^{20} = \begin{pmatrix} u_{22}^{12} & -1 & 0 \\ -1 & u_{22}^{22} & 0 \\ 0 & 0 & -\frac{u_{22}^{11}u_{22}^{12} - 2u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} \end{pmatrix}$$

$$h_1 3^{21} = \begin{pmatrix} u_{22}^{22} & 0 & -1 \\ 0 & -\frac{u_{22}^{11}u_{22}^{12} - 2u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} & 0 \\ -1 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 3^{22} = \begin{pmatrix} u_{22}^{11} & 0 & 0 \\ 0 & -\frac{\left(u_{22}^{11}\right)^2 - 2u_{22}^{11}u_{22}^{12} - u_{22}^{11}u_{22}^{22} + 4u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} & 1 \\ 0 & 1 & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 13 in family Z3_RR_diag-gauge with pattern S=[(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)] chart=2, comp 0:

$$\gamma_1^{(0,0)}3 = g_{22}$$

$$\gamma_1^{(0,1)}3 = -\frac{g_{20}u_{22}^{11} - 2g_{20}u_{22}^{12} + g_{20}u_{22}^{22} - g_{21}u_{22}^{11} + 2g_{21}u_{22}^{12} - g_{21}u_{22}^{22} - g_{22}u_{22}^{11} + 2g_{22}u_{22}^{12} - g_{22}u_{22}^{22} + \left(u_{22}^{11}\right)^2 - u_{22}^{11}u_{22}^{12} - u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}}$$

$$\gamma_1^{(0,2)}3 = -g_{20} + 2g_{22}$$

$$\gamma_1^{(1,0)}3 = \frac{g_{20}u_{22}^{11} - 2g_{20}u_{22}^{12} + g_{20}u_{22}^{22} - g_{21}u_{22}^{11} + 2g_{21}u_{22}^{12} - g_{21}u_{22}^{22} + g_{22}u_{22}^{11} - 2g_{22}u_{22}^{12} + g_{22}u_{22}^{22} + \left(u_{22}^{11}\right)^2 - u_{22}^{11}u_{22}^{12} - u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}}$$

$$\gamma_1^{(1,1)}3 = g_{22}$$

$$\gamma_1^{(1,2)}3 = -g_{21} + 2g_{22}$$

$$\gamma_1^{(2,0)}3 = g_{20}$$

$$\gamma_1^{(2,1)}3 = g_{21}$$

$$\gamma_1^{(2,2)}3 = g_{22}$$

Genuine Solution no. 14 for family Z3_RR_diag-gauge with pattern S=[(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)] chart=2, comp 1:

$$h_1 4^{00} = \begin{pmatrix} u_{22}^{22} & 0 & 1 \\ 0 & u_{22}^{11} & 0 \\ 1 & 0 & -\frac{\left(u_{22}^{11}\right)^2 - 2u_{22}^{11}u_{22}^{12} - u_{22}^{11}u_{22}^{22} + 4u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} \end{pmatrix}$$

$$h_1 4^{01} = \begin{pmatrix} -\frac{u_{22}^{11}u_{22}^{12} - 2u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} & 0 & 0 \\ 0 & u_{22}^{12} & 1 \\ 0 & 1 & u_{22}^{22} \end{pmatrix}$$

$$h_1 4^{02} = \begin{pmatrix} u_{22}^{12} & 1 & 0 \\ 1 & u_{22}^{22} & 0 \\ 0 & 0 & -\frac{u_{22}^{11}u_{22}^{12} - 2u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} \end{pmatrix}$$

$$h_1 4^{10} = \begin{pmatrix} -\frac{u_{22}^{11}u_{22}^{12} - 2u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} & 0 & 0 \\ 0 & u_{22}^{12} & 1 \\ 0 & 1 & u_{22}^{22} \end{pmatrix}$$

$$h_1 4^{11} = \begin{pmatrix} -\frac{\left(u_{22}^{11}\right)^2 - 2u_{22}^{11}u_{22}^{12} - u_{22}^{11}u_{22}^{22} + 4u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} & 1 & 0 \\ 1 & u_{22}^{22} & 0 \\ 0 & 0 & u_{22}^{11} \end{pmatrix}$$

$$h_1 4^{12} = \begin{pmatrix} u_{22}^{22} & 0 & 1 \\ 0 & -\frac{u_{22}^{11}u_{22}^{12} - 2u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} & 0 \\ 1 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 4^{20} = \begin{pmatrix} u_{22}^{12} & 1 & 0 \\ 1 & u_{22}^{22} & 0 \\ 0 & 0 & -\frac{u_{22}^{11}u_{22}^{12} - 2u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} \end{pmatrix}$$

$$h_1 4^{21} = \begin{pmatrix} u_{22}^{22} & 0 & 1 \\ 0 & -\frac{u_{22}^{11}u_{22}^{12} - 2u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} & 0 \\ 1 & 0 & u_{22}^{12} \end{pmatrix}$$

$$h_1 4^{22} = \begin{pmatrix} u_{22}^{11} & 0 & 0 \\ 0 & -\frac{\left(u_{22}^{11}\right)^2 - 2u_{22}^{11}u_{22}^{12} - u_{22}^{11}u_{22}^{22} + 4u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}} & 1 \\ 0 & 1 & u_{22}^{22} \end{pmatrix}$$

htilde entries for Solution no. 14 in family Z3_RR_diag-gauge with pattern S=[(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)] chart=2, comp 1:

$$\gamma_1^{(0,0)}4 = g_{22}$$

$$\gamma_1^{(0,1)}4 = -\frac{g_{20}u_{22}^{11} - 2g_{20}u_{22}^{12} + g_{20}u_{22}^{22} - g_{21}u_{22}^{11} + 2g_{21}u_{22}^{12} - g_{21}u_{22}^{22} - g_{22}u_{22}^{11} + 2g_{22}u_{22}^{12} - g_{22}u_{22}^{22} + \left(u_{22}^{11}\right)^2 - u_{22}^{11}u_{22}^{12} - u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}}$$

$$\gamma_1^{(0,2)}4 = -g_{20} + 2g_{22}$$

$$\gamma_1^{(1,0)}4 = \frac{g_{20}u_{22}^{11} - 2g_{20}u_{22}^{12} + g_{20}u_{22}^{22} - g_{21}u_{22}^{11} + 2g_{21}u_{22}^{12} - g_{21}u_{22}^{22} + g_{22}u_{22}^{11} - 2g_{22}u_{22}^{12} + g_{22}u_{22}^{22} + \left(u_{22}^{11}\right)^2 - u_{22}^{11}u_{22}^{12} - u_{22}^{11}u_{22}^{22} - 2\left(u_{22}^{12}\right)^2 + 5u_{22}^{12}u_{22}^{22} - 2\left(u_{22}^{22}\right)^2 - 1}{u_{22}^{11} - 2u_{22}^{12} + u_{22}^{22}}$$

$$\gamma_1^{(1,1)}4 = g_{22}$$

$$\gamma_1^{(1,2)}4 = -g_{21} + 2g_{22}$$

$$\gamma_1^{(2,0)}4 = g_{20}$$

$$\gamma_1^{(2,1)}4 = g_{21}$$

$$\gamma_1^{(2,2)}4 = g_{22}$$